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NileCAM81_CUOAGX



Datasheet

Revision 1.6
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1 Revision History

Rev	Date	Major Changes	Author
1.0	26-Jul-2022	Initial draft	Camera Product Team
1.1	09-Nov-2022	Power measurement updated	Camera Product Team
1.2	23-Jan-2023	Updated sensor optical form-format	Camera Product Team
1.3	30-Jan-2023	Updated Frame rate and modified ubuntu version	Camera Product Team
1.4	10-Feb-2023	Added Sync Support	Camera Product Team
1.5	24-May-2023	Updated Frame Rate	Camera Product Team
1.6	08-Jan-2024	Updated web and e-mail in the first page	Camera Product Team



2 Introduction

NileCAM81_CUOAGX is family of NileCAM cameras from e-con Systems. It is provided with high-speed unified serial interface that carries video data, control data and power in a single coaxial cable. It uses serializer-deserializer (SerDes) technology that allows single coaxial cable interface for the camera. With coaxial cable lengths of up to 15m, NileCAM81_CUOAGX offers greater flexibility for the system architects to place the cameras in mechanically challenging designs of new generation embedded, automotive and autonomous driving applications. The flexibility of coaxial cable and the ability to carry video data, control data and power in a single cable makes NileCAM81_CUOAGX fast (low latency), flexible and yet reliable.

NileCAM81_CUOAGX uses Maxim's Gigabit Multimedia Serial Link (GMSL2) interface technology for the serial interface to the camera. This serial interface uses coaxial cable that carries high-speed video data from the camera, bidirectional control data between the camera and host controller, and power supply for the camera. The power to the camera is supplied from the host processor through this coaxial cable. NileCAM81_CUOAGX embeds the Maxim's serializer which serializes the high-speed image data and transmits over coaxial cable. The serializer also supports the bidirectional control data communication between the camera and host controller. The on-board electronics of NileCAM81_CUOAGX serializer recovers the power supplied from the host controller without contaminating the data transfers.

NileCAM81_CUOAGX board has been designed and developed by e-con Systems for the NVIDIA® Jetson™ AGX Xavier and NVIDIA® Jetson™ AGX Orin development kits. NileCAM81_CUOAGX is an 8 MP GMSL camera contains a 1/1.7" AR0821 CMOS image sensor from onsemi™ along with on-board ISP. The coaxial cable is connected to Serializer board through a rugged FAKRA connector. NileCAM81_CUOAGX can be directly interfaced with Jetson™ AGX Xavier/Orin development kit through J509 connector.

The NVIDIA® Jetson™ AGX Xavier/Orin developer kit is a full-featured development platform for visual computing. It is ideal for applications requiring high computational performance in a low power envelope.

e-con Systems provides the sample applications that demonstrates the features of this camera. However, this camera can also be utilized by any V4L2 application.

This document describes about the features of NileCAM81_CUOAGX including the mechanical diagram.

3 Disclaimer

The specifications and features of NileCAM81_CUOAGX camera board are provided here as reference only and e-con Systems reserves the right to edit/modify this document without any prior intimation of whatsoever.



4 Description

The NileCAM81_CUOAGX board is designed to interface up to six 8 MP custom lens camera modules with the Jetson™ AGX Xavier/Orin development kits. NileCAM81_CUOAGX is a three-board solution and has two parts. The first part has two boards: one is standard NileCAM81_CUOAGX camera module that has 1/1.7" AR0821 CMOS image sensor from onsemi™ along with Image Signal Processor (ISP) and the other one is Serializer board. The second part is Deserializer board which is used to interface with Jetson™ Xavier/Orin development kit.

NileCAM81_CUOAGX is a multi-board solution, which has three boards as follows:

- e-CAM82_CUMI0821_MOD_H02R1 (Camera Module Board)
- e-CAM22_CUMI_SER (Serializer Board)
- e-CAM_CUOAGX_DESER_6H01R1(Deserializer Board)

The camera module is a small, low-power, high performance 8 MP camera with a built-in ISP. It is based on AR0821 CMOS image sensor from onsemi™. The AR0821 is a 1/1.7" optical form-factor, CMOS image sensor with electronic rolling shutter.

The following figures show the Camera Module board, Serializer board and Deserializer board.

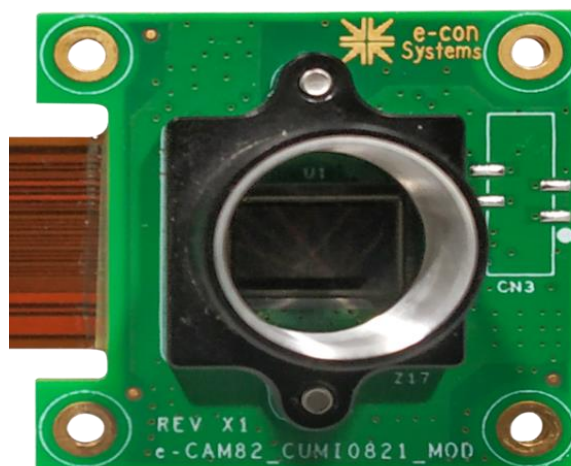


Figure 1: NileCAM81_CUOAGX Camera Module Front View





Figure 2: NileCAM81_CUOAGX Serializer Board

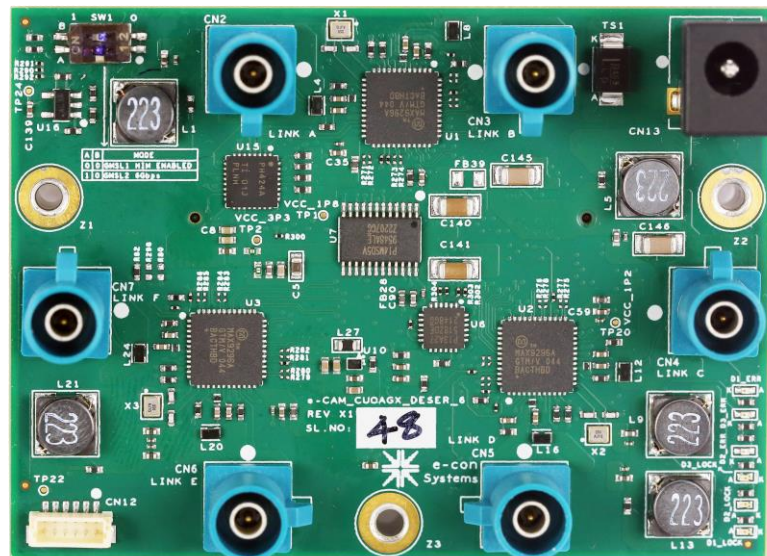


Figure 3: NileCAM81_CUOAGX Deserializer Board

The camera board is based on AR0821, a 1/1.7" CMOS image sensor from onsemi™ along with ISP. The camera is available with M12 metal lens holder allowing customers to choose the lens as per their requirements. The Serializer board is based on Maxim GMSL serializer and has all the circuitry required for bidirectional data communication, power and data separation and so on. FAKRA connectors are used for the coaxial cable interface. The Deserializer board needs external power supply, which supplies the power to the Serializer board over coaxial cable using the FAKRA connector. The Deserializer board has six Fakra connectors, through which you can connect six cameras at the same time.



NileCAM81_CUOAGX has six different variants which are listed in following table.

Part Number	Deserializer Board	Serializer Board	Module Board	Coaxial Cable	Lens
NileCAM81_CUOAGX_1H01R1	1	1	1	1 (3m cable)	1
NileCAM81_CUOAGX_1H02R1	1	1	1	1 (15m cable)	1
NileCAM81_CUOAGX_4H01R1	1	4	4	1 (3m cable)	4
NileCAM81_CUOAGX_4H02R1	1	4	4	1 (15m cable)	4
NileCAM81_CUOAGX_6H01R1	1	6	6	6 (3m cable)	6
NileCAM81_CUOAGX_6H02R1	1	6	6	6 (15m cable)	6

Table 1: NileCAM81_CUOAGX Variants

Note: The default variants of NileCAM81_CUOAGX are given in the above table. For customization on different number of camera modules, you can write to techsupport@e-consystems.com

NileCAM81_CUOAGX operates in two modes as follows:

- Asynchronous Mode
- Synchronous Mode

4.1 Asynchronous Mode

The asynchronous mode is the normal streaming mode. In this mode, all six cameras can be controlled individually.

The below table list the supported frame rates in asynchronous mode.

Resolution	DAY HDR Frame Rate (fps)	NIGHT HDR Frame Rate (fps)	Linear Frame Rate(fps)
1280 x 720	30	60	60
1920 x 1080	30	60	60
3840 x 2160	16	16	16

Table 2: Maximum Frame Rates in Asynchronous Mode

4.2 Synchronous Mode

The synchronous mode is the special feature of NileCAM81_CUOAGX board. The NileCAM81_CUOAGX board contains on-board Pulse Width Modulation (PWM) generator circuit to provide the necessary trigger signal for synchronous mode. The below table list the supported frame rates in synchronous mode.

Resolution	DAY HDR Frame Rate (fps)	NIGHT HDR Frame Rate (fps)	Linear Frame Rate(fps)
1280 x 720	30	60	60
1920 x 1080	30	60	60
3840 x 2160	15	16	15



Table 3: Maximum Frame Rates in Synchronous Mode

NileCAM81_CUOAGX synchronize all the frames of camera according to the provided PWM pulse as trigger. This PWM trigger pulses can be provided internally (Internal Trigger Mode) or externally (External Trigger Mode).

In Internal Trigger Mode, NileCAM81_CUOAGX contains on-board PWM generator circuit to provide the necessary trigger signal for frame synchronization. The output from each camera is frame synchronized. The overall tolerance for the generated PWM signal frequency is 1%. e-con Systems presently supports 30 Hz PWM pulse in Internal Trigger Mode. For customization of PWM frequency, please write to techsupport@e-consystems.com.

In External Trigger Mode, NileCAM81_CUOAGX board has an I/O Header (CN12), through which you can provide PWM pulse of any frequency (less than 60Hz) to synchronize all camera frames. The external trigger signal IO voltage is 3.3V. The Internal or External Trigger Mode selected through software. To connect with external trigger refer the CN12 connector pin outs section.

Note: Synchronization for 4k resolution can only be provided by external trigger, since the minimum internal trigger is 30Hz. So, we cannot sync 15Hz signal through on board PWM generator.

4.3 Features

The features of NileCAM81_CUOAGX are as follows:

- Multi-board solution
- AR0821 8MP camera sensor with uncompressed UYVY format
- High Dynamic Range Imaging
- Compatible with Jetson AGX Xavier™ /Orin™ development kit
- Standard M12 lens holder for use with customized optics or lenses for various applications
- Asynchronous and synchronous modes.
- Light weight, versatile, and portable design
- Imaging applications
 - 8 MP CMOS image sensor with 1/1.7" optical form-factor
 - Still capture supported resolution: HD (1280 x 720), FHD (1920 x 1080), 4K (3840 x 2160)
 - Video streaming supported resolution: HD (1280 x 720), FHD (1920 x 1080), 4K (3840 x 2160)
- Maximum power consumed: 1.13W
- Operating temperature range: -30°C to 70°C
- RoHS compliant



4.4 GMSL Mode Selecting Switch

NileCAM81_CUOAGX is backward compatible with our GMSL1 cameras (NileCAM_30). **SW1** switch is used for selecting GMSL 1 or GMSL2 operations. The position of switch is shown in the following figure.

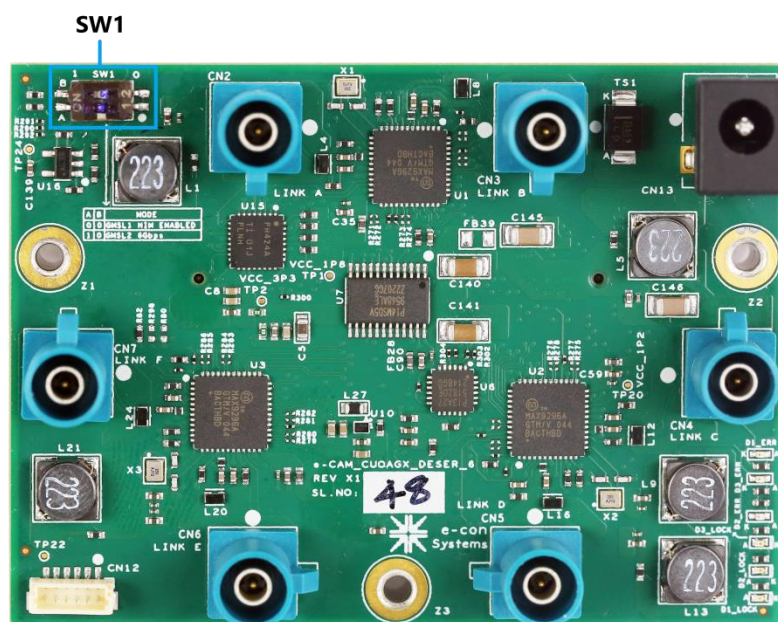


Figure 4: Position of SW1 Switch

The following table describes the different modes in the switch.

A	B	Mode
0	0	GMSL 1 HIM Enabled
0	1	GMSL 2 3Gbps
1	0	GMSL 2 6Gbps
1	1	GMSL 1 HIM Disabled

Table 4: Modes in Switch

Note: For NileCAM81_CUOAGX product, GMSL2 6Gbps mode is selected by default. For NileCAM81_CUOAGX to operating in GMSL1 mode, please write to techsupport@e-consystems.com

5 Key Specifications

The following table lists the key specifications of NileCAM81_CUOAGX.

Description	Specification
Video format	UYVY
Maximum image resolution	3840 x 2160
Supported OS	Ubuntu 20.04



Table 5: Key Specifications of NileCAM81_CUOAGX

5.1 CMOS Image Sensor Specifications

The following table lists the specifications of the CMOS image sensor used in this NileCAM81_CUOAGX camera board.

Sensor Specification	
Type/Optical size	1/1.7" Optical format CMOS image sensor
Resolution	8 MP
Image Format	UYVY-Electronic rolling shutter
Pixel size	2.1 μm
Sensor active area	3848H x 2168V

Table 6: CMOS Image Sensor Specification

For more information about the AR0821 sensor or for Datasheet, please contact onsemi™.

6 Pin Description

The following sections describe the pins descriptions of the serializer board and deserializer board connectors.

6.1 Serializer Board Connectors

The serializer board has two on-board connectors for user interface. The following section describes the pinouts and connector locations.

6.1.1 Serializer to Module Mating Connector (CN1)

NileCAM81_CUOAGX Serializer board has single dual rows, 26-pin Samtec connectors. CN1 connectors are used for mating with module board. The connector on Serializer board is shown in the following figure.

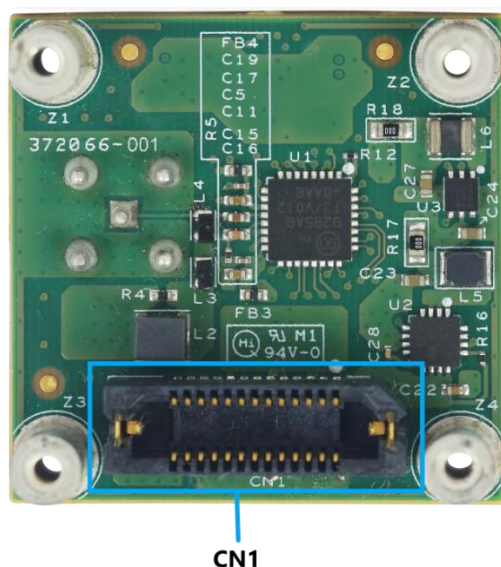


Figure 5: Connectors on Serializer Board

The CN1 pin numbers, signal names, pin types and their description from serializer board perspective are listed in following table.

CN1 Pin No	Signal Name	Pin Type	Description
1	MIPI_CLK_N	INPUT	MIPI Clock Lane Differential Pair -
2	MIPI_DATA0_N	INPUT	MIPI Data Lane 0 Differential Pair -
3	MIPI_CLK_P	INPUT	MIPI Clock Lane Differential Pair +
4	MIPI_DATA0_P	INPUT	MIPI Data Lane 0 Differential Pair +
5	GND	POWER	Ground signal for digital and analog
6	GND	POWER	Ground signal for digital and analog
7	RSVD	-	Reserved
8	CAM_I2C_SCL	OUTPUT	1.8V IO voltage I2C Clock signal with internal pull-up 4.7K
9	RSVD	-	Reserved
10	CAM_I2C_SDA	BI-DIR	1.8V IO voltage I2C Data signal with internal pull-up 4.7K
11	GND	POWER	Ground signal for digital and analog
12	CAM_RESET	OUTPUT	1.8V IO reset signal for camera module
13	RSVD	-	Reserved
14	CAM_BOOT	OUTPUT	1.8V IO Boot signal for camera module
15	RSVD	-	Reserved
16	GND	POWER	Ground signal for digital and analog
17	GND	POWER	Ground signal for digital and analog
18	RSVD	-	Reserved
19	MIPI_DATA1_N	OUTPUT	MIPI Data Lane 1 Differential Pair -
20	RSVD	-	Reserved
21	MIPI_DATA1_P	OUTPUT	MIPI Data Lane 1 Differential Pair +
22	RSVD	-	Reserved
23	GND	POWER	Ground signal for digital and analog
24	Trigger	OUTPUT	1.8V IO trigger signal for camera
25	CAM_FLASH	INPUT	1.8V IO camera Flash
26	VCC_3P3	POWER	3.3V Power supply for camera boards

Table 7: CN1 Pin Descriptions

6.1.2 FAKRA Straight Plug (Coax Connector) Pin Description (CN2)

The connector on serializer board is shown in the following figure.



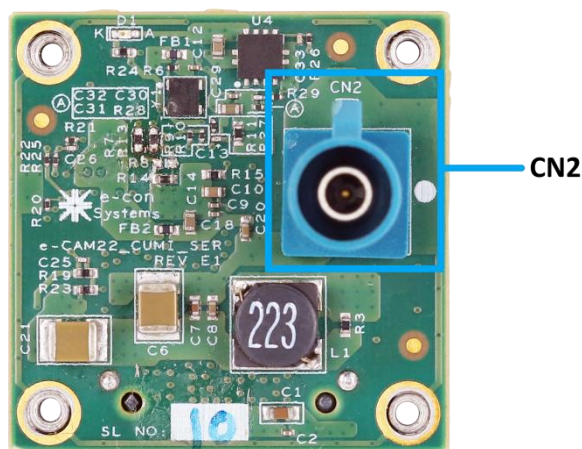


Figure 6: Connector on Serializer Board

The serializer board has a high frequency FAKRA straight plug (CN2) with neutral coding, to mate with coaxial cable. The same FAKRA straight plug and pinouts are used in the Deserializer board.

The following table lists the pinouts details of FAKRA plug.

Pin No	Signal Name	Pin Type	Description
1	Coax	INPUT/ OUTPUT	Coaxial signal that carries both power and bidirectional data
2	GND	POWER	Ground
3	GND		Ground
4	GND		Ground
5	GND		Ground

Table 8: FAKRA Plug Pinouts Details

6.2 Deserializer Board Connectors

The Deserializer board has six FAKRA connectors (CN2-CN7) power supply connector (CN13) and External trigger connector (CN12). The following figure shows the location of the connectors.



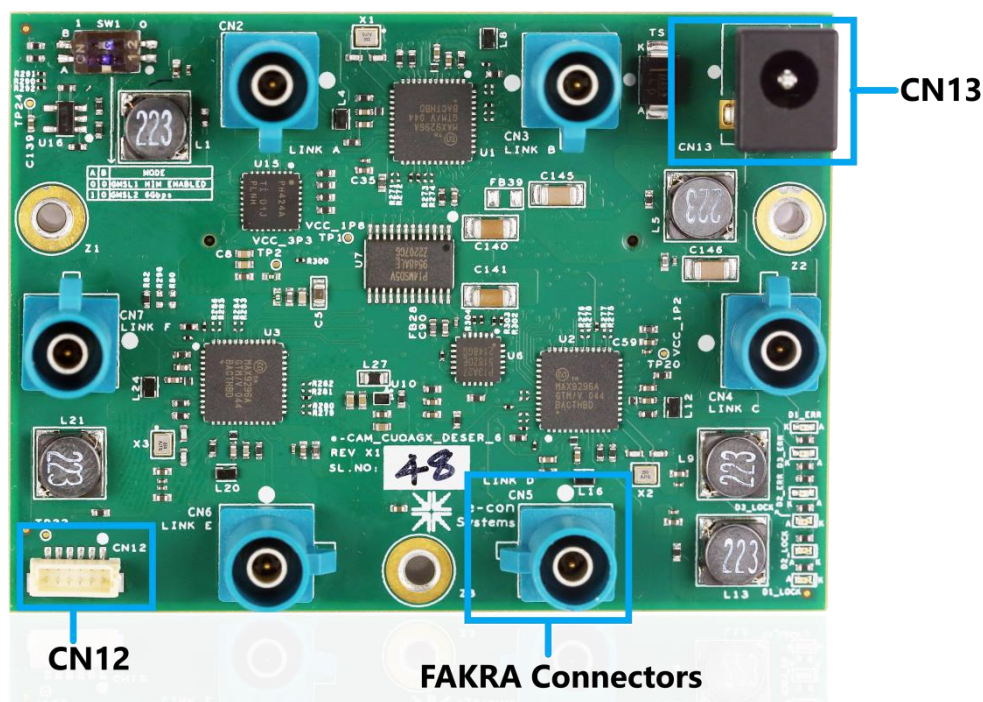


Figure 7: Connectors on Deserializer Board

6.2.1 Power Supply Connector (CN13)

This connector is used for supplying power for NileCAM81_CUOAGX camera module through POC. This DC jack connector has internal diameter 1.75mm and outer diameter 4.75mm. e-con Systems provides the mating power supply adaptor with 12V 2A.

The following table lists the CN13 connector pin-out details.

Pin No	Signal Name	Pin Type	Description
1	VCC_POC	POWER	Power for camera Modules. Supports 5V to 12V.
2	GND	POWER	Ground
3	GND	POWER	Ground

Table 9: CN3 Connector Pin-out Details

Note: Kindly use e-con Systems provided adaptor or make ensure your power adaptor supports 24W power.

6.2.2 FAKRA Connectors (CN2-CN7)

NileCAM81_CUOAGX deserializer board has six FAKRA connectors for interfacing with six cameras via serializer board. This connector is used for carrying power to camera modules



and receiving high speed data from camera module through POC. The following table lists the FAKRA plug pin-out details.

Pin No	Signal Name	Pin Type	Description
1	Coax	INPUT/ OUTPUT	Coaxial signal that carries both power and bidirectional data
2	GND	POWER	Ground
3	GND		Ground
4	GND		Ground
5	GND		Ground

Table 10: FAKRA Plug Pin-outs Details

6.2.3 External Trigger connector (CN12)

The external trigger connector is used for providing external trigger for NileCAM81_CUOAGX and an external GPIO option for camera control is available. The following table lists the external trigger connector details.

Pin No	Signal	Pin Type	Description
1	VCC_3P3	POWER	3.3V power for external circuitry
2	EXT_TRIG	INPUT	3.3V IO PWM input signal for cameras. All connected cameras frame is synced to this PWM frequency.
3	D1_MFP6	BI-DIR	3.3V IO GPIO signal for LINK A & LINK B cameras
4	D2_MFP6	BI-DIR	3.3V IOGPIO signal for LINK C & LINK D cameras
5	D3_MFP6	BI-DIR	3.3V IOGPIO signal for LINK E & LINK F cameras
6	GND	POWER	Ground

Table 11: CN12 Connector Pin-outs Details

6.3 Connector Part Numbers

The following table lists the connectors and cables used in NileCAM81_CUOAGX and its compatible mating connector and cable.

Board	Description	Connector	Manufacturer	Part Number
Camera Module	NileCAM81_CUOAGX MOD board dual row connector (CN2) for mating with NileCAM81_CUOAGX serializer board (CN1)	CONN Board to Board Header Center Strip Contacts P-0.80mm 26Pos Dual Row Vertical SMT (CN2)	Samtec	ERM8-013-03.0-L-DV-L-K-TR



Serializer Board	NileCAM81_CUOAGX serializer board dual row connector (CN1) for mating with NileCAM81_CUOAGX camera module (CN2)	CONN Board to Board Receptacle Outer Shroud Contacts P-0.80mm 26Pos Dual Row Vertical SMT (CN1)	Samtec	ERF8-013-05.0-L-DV-L-K-TR
	NileCAM81_CUOAGX serializer board CONN FAKRA Plug to connect Coaxial cable to Vertical Through Hole (CN2)	CONN FAKRA Plug Receptacle 5Pos 50Ohm 6GHz Vertical Through Hole (CN2)	Rosenberger	59S10H-400T5-Z
Deserializer Board	NileCAM81_CUOAGX Deserializer board mating connector with Jetson AGX Xavier/Orin (CN9)	CONN Board to Board Outer Shroud Contacts P-0.50mm 120Pos Dual Row Vertical SMT (CN9)	Samtec	QTH-060-02-H-D-A
	NileCAM81_CUOAGX Deserializer board Power supply plug for GMSL support (CN13)	CONN Jack Power Male 2 Conductors, 3 Contacts 24VDC 5A Right Angle Through Hole (CN13)	CUI Inc	PJ-042-SMT
	NileCAM81_CUOAGX Deserializer board External trigger connector (CN12)	CONN Header Male P-1mm Shrouded 6Pos Vertical SMT (CN12)	JST Sales America Inc	BM06B-SRSS-TB(LF)(S N)
	NileCAM81_CUOAGX Deserializer board dual row connector (CN1) for mating NileCAM81_CUOAGX camera module	CONN FAKRA Plug Receptacle 5Pos 50Ohm 6GHz Vertical Through Hole (CN2, CN3, CN4, CN5, CN6, CN7)	Rosenberger	59S10H-400T5-Z

Table 12: NileCAM81_CUOAGX Connector Details

7 Electrical Specification

This section lists the electrical specification and recommended operating conditions of NileCAM81_CUOAGX.

7.1 Recommended Operating Condition

NileCAM81_CUOAGX requires two power sources for normal operation. One is for deserializer board and the other is for camera module plus serializer boards. NileCAM81_CUOAGX uses Orin camera expansion connector (J509) power for powering deserializer board and CN13 connector power for powering camera modules.



7.1.1 Power Consumption (from Orin) of Deserializer Board

NileCAM81_CUOAGX deserializer board uses only 1.8V and 3.3V power supplies from Orin camera expansion connector where else 2.8V and 5V power supplies are unused.

No of Cameras	Parameter	Typical Operating Voltage	Typical Power Consumption (W)
One	Input voltage	3.3V	0.0132
One	Input voltage	1.8V	1.0548
Six	Input voltage	3.3V	0.0165
Six	Input voltage	1.8V	2.196

Table 13: NileCAM81_CUOAGX Deserializer board power consumption

Note: The values described in this section are measured in e-con Systems lab and this can be used as reference only. The current measurements are typical values and are subject to change for different camera boards under different conditions. However, these values can be taken as a reference for power estimation and power supply design.

7.2 Power Consumption of Serializer and Module Boards

For Camera modules and serializer boards, NileCAM81_CUOAGX uses power from CN13 connector, which supports 5 to 15V power supply. By default, e-con provides 12V 2A power adapter. The power consumption details for NileCAM81_CUOAGX in one, four and six cameras in synchronous and asynchronous modes are listed below.

In Synchronous Mode power consumption:

The following table lists the power consumption of NileCAM81_CUOAGX in Synchronous Mode.

Sl.No	Mode	Resolution	Frame rate	Solo	Quad	Hex
1	Linear	1280 x 720	60	2.076	8.34	9.804
2		1920 x 1080	60	2.1	8.436	12.756
3		3840 x 2160	15	1.824	7.332	9.96
4	HDR	1280 x 720	30	2.448	7.164	9.72
5		1920 x 1080	30	2.484	7.488	9.828
6		3840 x 2160	15	2.016	7.5	10.944

Table 14: Power Consumption Details of NileCAM81_CUOAGX in Synchronous Mode

In Asynchronous Mode power consumption:

The following table lists the power consumption of NileCAM81_CUOAGX in Asynchronous Mode.



Sl.No	Mode	Resolution	Frame rate	Solo	Quad	Hex
1	Linear	1280 x 720	60	2.088	8.376	12.384
2		1920 x 1080	60	2.124	8.448	12.588
3		3840 x 2160	16	1.872	7.716	12.276
4	HDR	1280 x 720	30	2.52	7.416	13.008
5		1920 x 1080	30	2.556	7.572	13.044
6		3840 x 2160	16	2.064	7.908	12.588

Table 15: Power Consumption Details of NileCAM81_CUOAGX in Synchronous Mode

Note: These values are measured in e-con Systems lab, and this can be used as reference only. The current measurements are typical values and are subject to change for different camera boards under different conditions. However, these values can be taken as a reference for power estimation and power supply design.

NileCAM81_CUOAGX does not require any power sequence since it required only 3.3V power supply for operation.

7.3 Operating Temperature Range

The following table lists the operating temperature range of NileCAM81_CUOAGX.

Parameter Description	Temperature Range
Operating Temperature Range ¹	-30°C to 70°C

Table 16: Operating Temperature Range

¹This is the maximum temperature range up to which the camera sensor can be operated.

Note: The default lens supplied with this camera has an operating range of -30°C to 75°C. You can choose wider operating temperature lens as per your requirements. You are advised to make necessary arrangements on the products to dissipate the heat generated in the module to maintain the operating temperature below 70°C

8 Mechanical Specification

NileCAM81_CUOAGX is a multi-board solution. The camera module and serializer boards each of size is 30 mm x 30 mm. The dimension of deserializer board is 75 mmx55 mm.

8.1 Camera Module Board Dimensions

The camera board of NileCAM81_CUOAGX is a two-board solution connected with rigid flex cable each board having 30 mm x 30 mm in dimension. The front and rear views of the NileCAM81_CUOAGX module board with its dimensions are shown in the following figures.



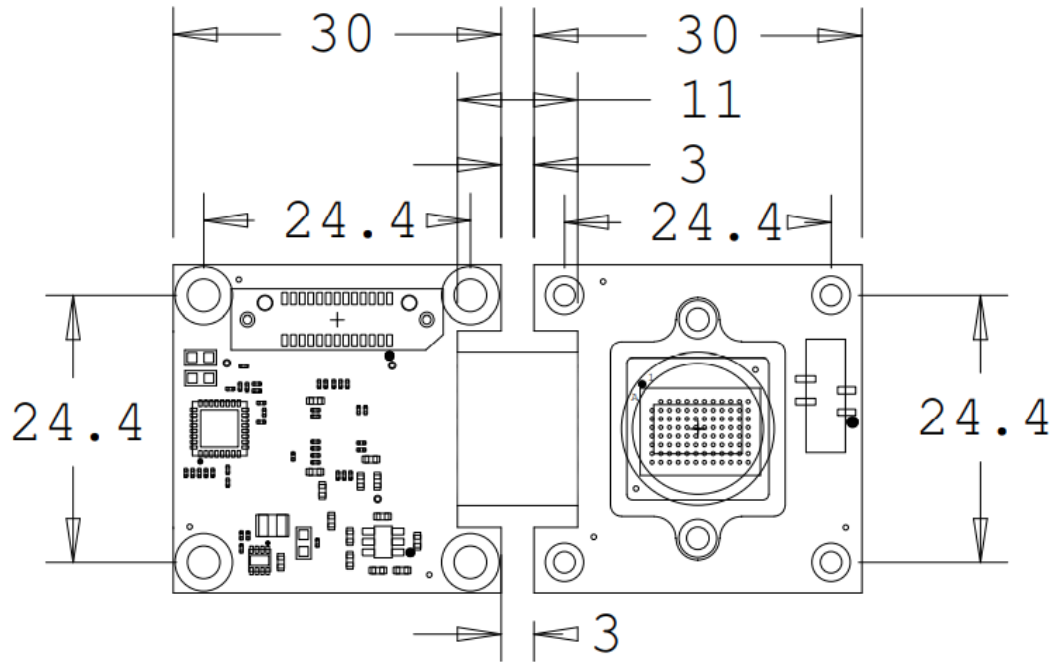


Figure 8: Front View of NileCAM81_CUOAGX Module Board Mechanical Dimensions

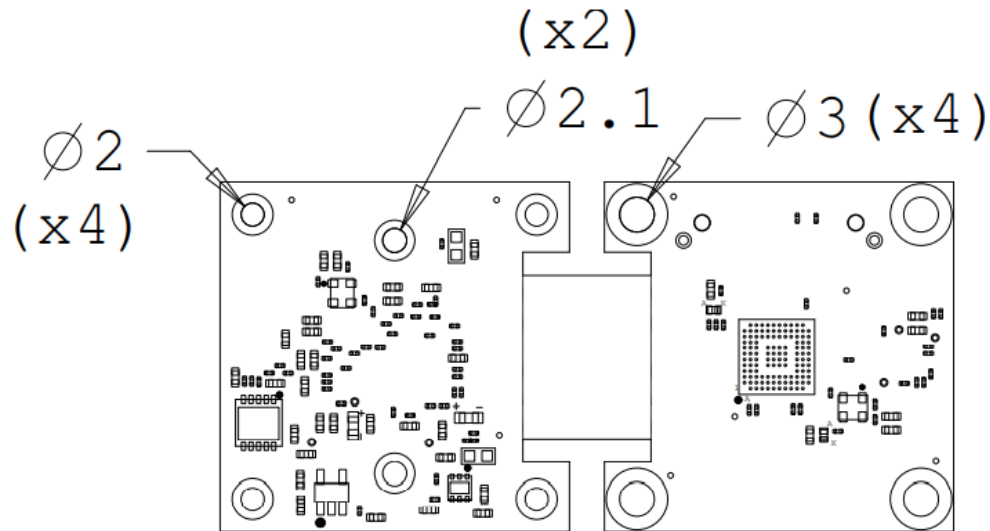


Figure 9: Rear View of NileCAM81_CUOAGX Module Board Mechanical Dimensions

Note: NileCAM81_CUOAGX module board is two board solution connected with rigid flex cable. However, it will be given as folded condition.

Note:

1. All dimensions are in mm.
2. Board tolerance will be of +/- 0.2 mm



8.2 NileCAM81_CUOAGX Serializer Board Dimensions

The front and rear views of NileCAM81_CUOAGX serializer board with mechanical details is shown in the following figures.

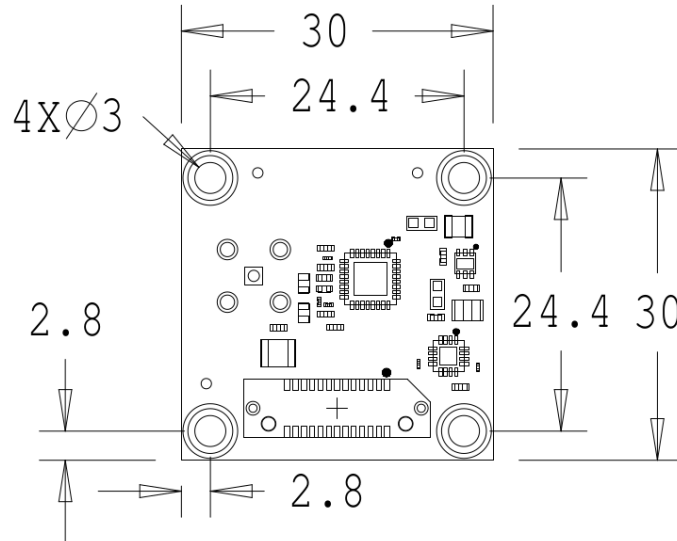


Figure 10: Front View of NileCAM81_CUOAGX Serializer Board Mechanical Dimensions

Note:

1. All dimensions are in mm.
2. Board tolerance will be of +/- 0.2 mm

8.3 Deserializer Board Dimensions

The front and rear views of NileCAM81_CUOAGX deserializer board with mechanical details are shown in the following figures.



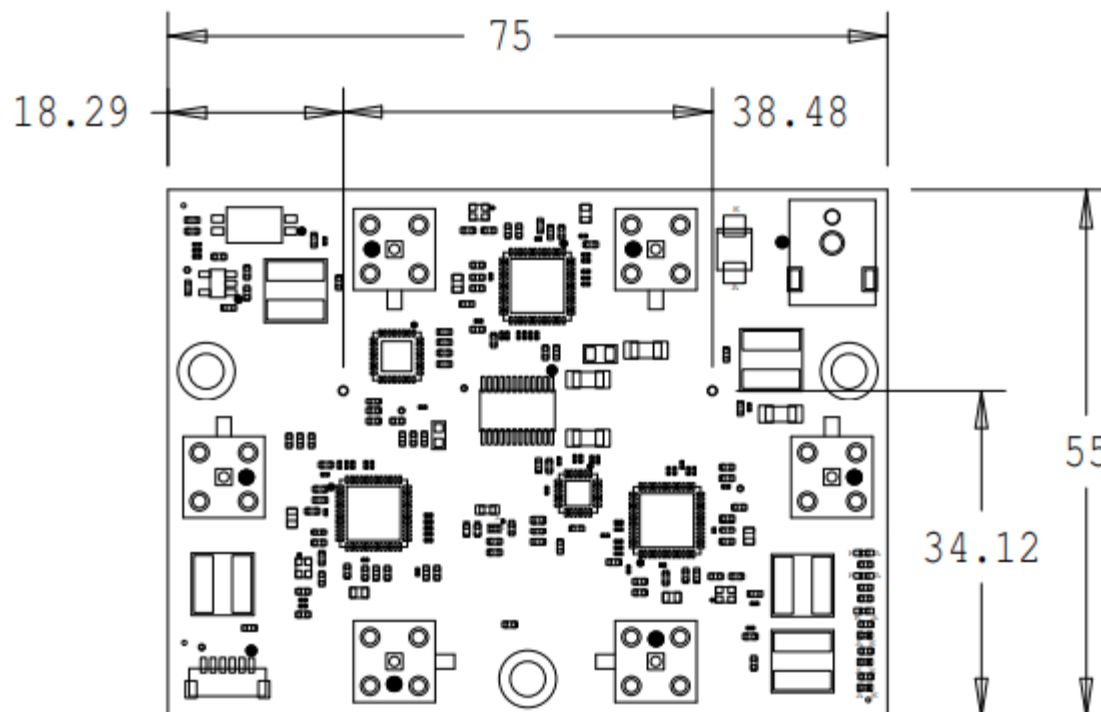


Figure 11: NileCAM81_CUOAGX Deserializer Board Front View

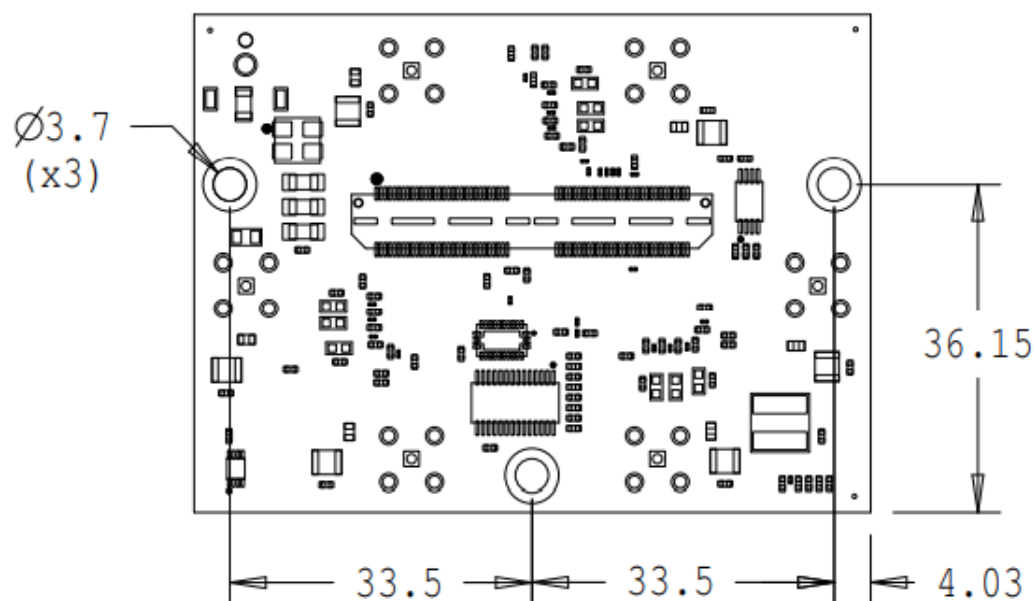


Figure 12: NileCAM81_CUOAGX Deserializer Board Rear View

Note:

1. All dimensions are in mm.
2. Board tolerance will be of ± 0.2 mm



8.4 Lens Holder Dimensions

The following figure shows the dimension details of S-Mount lens holder.

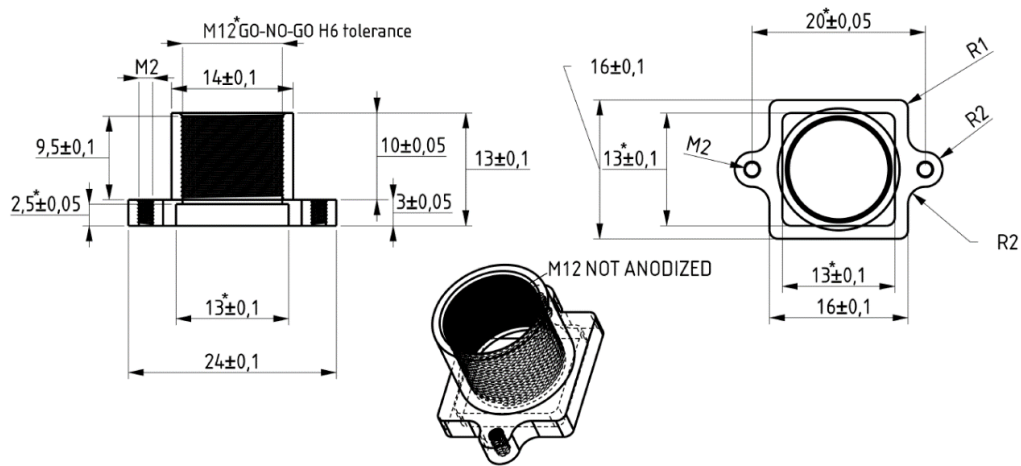


Figure 13: S-Mount Holder Outline Dimension

Note: All dimensions are in mm.



Support

Contact Us

If you need any support on NileCAM81_CUOAGX product, please contact us using the Live Chat option available on our website - <https://www.e-consystems.com/>

Creating a Ticket

If you need to create a ticket for any type of issue, please visit the ticketing page on our website - <https://www.e-consystems.com/create-ticket.asp>

RMA

To know about our Return Material Authorization (RMA) policy, please visit the RMA Policy page on our website - <https://www.e-consystems.com/RMA-Policy.asp>

General Product Warranty Terms

To know about our General Product Warranty Terms, please visit the General Warranty Terms page on our website - <https://www.e-consystems.com/warranty.asp>

